

COMP7103 Assignment 1

Suggested Solution

Question 1 Data preprocessing and distance measure

a) Cosine similarities:

Records	Similarity	Rank
R_1 vs. R_2	0.98615	4
R_2 vs. R_3	0.98527	5
R_3 vs. R_4	0.99910	3
R_4 vs. R_5	0.99990	1
R_5 vs. R_6	0.99922	2

R_4 and R_5 are the most similar.

b) $\mu_{Zone4} = 115.17, \mu_{Zone5} = 127.33$

$s_{Zone4} = 7.6790, s_{Zone5} = 2.4221$

$coVariance(Zone4, Zone5) = -16.867$

$corr(Zone4, Zone5) = -0.9068$

The two attributes are negatively correlated.

Question 2 Metric Axioms

Positivity is always satisfied as all costs are greater than zero. $d(p, q) = 0$ only if two sequences are identical.

Symmetry will not be satisfied if $C_i \neq C_d$. Consider when $p_2 = \boxed{i} \boxed{\text{love}} \boxed{\text{you}}$ and $q_2 = \boxed{\text{love}} \boxed{\text{you}}$, then $d(p_2, q_2) = C_d$ and $d(q_2, p_2) = C_i$, $d(p_2, q_2) \neq d(q_2, p_2)$ if $C_i \neq C_d$.

Triangle Inequality will always be satisfied. Given any sequences p_3, q_3, r_3 , $d(p_3, r_3)$ is at most $d(p_3, q_3) + d(q_3, r_3)$, as the edit distance considers all possible ways and take the minimum.

Question 3 Decision Tree Classifier

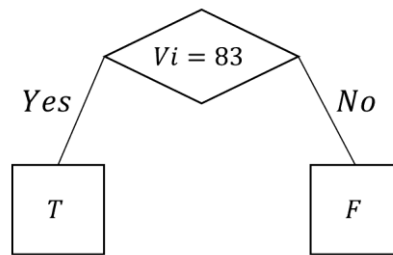
a) Splitting at V_i

Split point	23.5		42.5		51.5		63.5		83		107		150		194		209		243		378.5	
	<	≥	<	≥	<	≥	<	≥	<	≥	<	≥	<	≥	<	≥	<	≥	<	≥	<	≥
T	1	5	1	5	1	6	3	3	5	1	5	1	5	1	6	0	6	0	6	0	6	0
F	0	8	1	7	2	6	2	6	2	6	3	5	4	4	4	4	5	3	6	2	7	1
GINI	0.440		0.488		0.485		0.457		0.327		0.387		0.432		0.343		0.390		0.429		0.462	

Splitting at I_m

Split point	0.5		1.5		4		6.5		7.5		10		12.5		14.5	
	<	≥	<	≥	<	≥	<	≥	<	≥	<	≥	<	≥	<	≥
T	0	6	1	5	1	5	2	4	3	3	4	2	5	1	5	1
F	1	7	1	7	3	5	4	4	5	3	6	2	6	2	7	1
GINI	0.462		0.488		0.464		0.476		0.482		0.486		0.485		0.488	

Best split point is $Vi = 83$. Decision stump:



b) Discretized dataset:

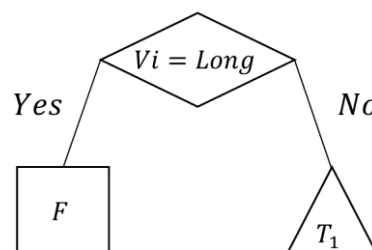
Records	Vi	Im	Class = T	Class = F
1, 3	Short	Low	1	1
2, 4, 5	Short	High	2	1
6, 7, 8	Mid	Low	2	1
9, 10	Mid	High	1	1
11, 12, 14	Long	Low	0	3
13	Long	High	0	1

$$P(\text{Class} = T) = \frac{6}{14}, P(\text{Class} = F) = \frac{8}{14}$$

$$\text{Entropy of the training set} = -\left(\frac{6}{14}\right)\log_2\left(\frac{6}{14}\right) - \left(\frac{8}{14}\right)\log_2\left(\frac{8}{14}\right) = 0.9852$$

Split	Entropy	Information gain
$Vi = \text{Short}$	$\frac{5}{14}\left(-\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)\right) + \frac{9}{14}\left(-\frac{3}{9}\log_2\left(\frac{3}{9}\right) - \frac{6}{9}\log_2\left(\frac{6}{9}\right)\right) = 0.9371$	$0.98525 - 0.9371 = 0.0481$
$Vi = \text{Long}$	$\frac{10}{14}\left(-\frac{6}{10}\log_2\left(\frac{6}{10}\right) - \frac{4}{10}\log_2\left(\frac{4}{10}\right)\right) + \frac{4}{14}(0) = 0.6935$	$0.9852 - 0.6935 = 0.2917$
$Im = \text{Low}$	$\frac{8}{14}\left(-\frac{3}{8}\log_2\left(\frac{3}{8}\right) - \frac{5}{8}\log_2\left(\frac{5}{8}\right)\right) + \frac{6}{14}\left(-\frac{3}{6}\log_2\left(\frac{3}{6}\right) - \frac{3}{6}\log_2\left(\frac{3}{6}\right)\right) = 0.9740$	$0.9852 - 0.9740 = 0.0113$

Best split is $Vi = \text{Long}$, tree is now:



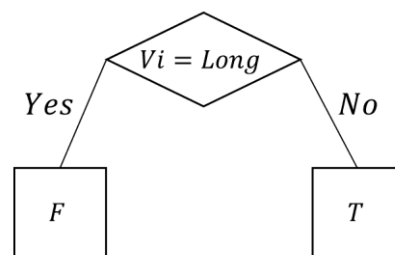
Consider subtree T_1 :

Records	V_i	I_m	Class = T	Class = F
1, 3	Short	Low	1	1
2, 4, 5	Short	High	2	1
6, 7, 8	Mid	Low	2	1
9, 10	Mid	High	1	1

Entropy of the training samples in $T_1 = -\left(\frac{6}{10}\right)\log_2\left(\frac{6}{10}\right) - \left(\frac{4}{10}\right)\log_2\left(\frac{4}{10}\right) = 0.9710$

Split	Entropy	Information gain
$V_i = \text{Short}$	$\frac{5}{10}\left(-\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)\right) + \frac{5}{10}\left(-\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)\right) = 0.9710$	$0.9710 - 0.9710 = 0$
$I_m = \text{Low}$	$\frac{5}{10}\left(-\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)\right) + \frac{5}{10}\left(-\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)\right) = 0.9710$	$0.9710 - 0.9710 = 0$

As information gain < 0.1 , we stop splitting. The decision tree becomes



c) Calculates split condition:

Record	V_i	I_m	$V_i > 10I_m$	Predicted Class	Actual Class
1	5	7	No	T	F
2	42	8	No	F	F
3	43	2	Yes	F	T
4	60	8	No	T	F
5	60	12	No	T	F
6	67	1	Yes	T	T
7	67	6	Yes	T	T
8	99	2	Yes	F	T
9	115	16	No	T	F
10	185	16	Yes	T	T
11	203	0	Yes	F	T
12	215	6	Yes	F	T
13	271	13	Yes	F	T
14	486	7	Yes	F	T

Confusion matrix:

		Predicted	
		<i>T</i>	<i>F</i>
Actual	<i>T</i>	3	3
	<i>F</i>	2	6

$$\text{Precision} = \frac{3}{5} = 0.6, \text{Recall} = \frac{3}{6} = \frac{1}{2} = 0.5, \text{F-measure} = \frac{2 \cdot 0.6 \cdot 0.5}{0.6 + 0.5} = \frac{6}{11} = 0.5455$$

Question 4 Classification in Weka

- a) @RELATION "Gallstone"
 @ATTRIBUTE Gender {Male,Female}
 @ATTRIBUTE Comorbidity {Yes,No}
 @ATTRIBUTE CAD {Yes,No}
 @ATTRIBUTE Hypothyroidism {No,Yes}
 @ATTRIBUTE Hyperlipidemia {Yes,No}
 @ATTRIBUTE DMx {Yes,No}
 @ATTRIBUTE HFA {Yes,No}
 @ATTRIBUTE class {Yes,No}

- b) (Note: result may vary)

The final model:

```
J48 pruned tree
-----
-0.691Gender=Female+0.467Hyperlipidemia=No-0.421Hypothyroidism=Yes-0.302CAD=No-0.154DMx=No... <= -0.265488: Yes (155.0/64.0)
-0.691Gender=Female+0.467Hyperlipidemia=No-0.421Hypothyroidism=Yes-0.302CAD=No-0.154DMx=No... > -0.265488
| -0.645Hypothyroidism=Yes-0.526Hyperlipidemia=No-0.386HFA=No+0.358CAD=No+0.136Comorbidity=No... <= 0.209001: No (82.0/26.0)
| -0.645Hypothyroidism=Yes-0.526Hyperlipidemia=No-0.386HFA=No+0.358CAD=No+0.136Comorbidity=No... > 0.209001
| | -0.664Comorbidity=No-0.556DMx=No-0.34HFA=No-0.246CAD=No-0.199Hyperlipidemia=No... <= 0.958831: No (63.0/30.0)
| | -0.664Comorbidity=No-0.556DMx=No-0.34HFA=No-0.246CAD=No-0.199Hyperlipidemia=No... > 0.958831: Yes (19.0/8.0)
```

The confusion matrix:

```
=== Confusion Matrix ===
   a  b  <-- classified as
95 63 |  a = Yes
80 81 |  b = No
```

Algorithms used:

- **Principle component analysis** is first used to select 5 attributes (after transformation).
- **CVParameterSelection** is then used to select a value for “C” for the J48 algorithm.
- **J48 algorithm** is used with the selected “C” value to build the final decision tree.

Question 5 Classification in Python

a) Code segment:

```
import pandas as pd
from sklearn import tree
from sklearn.model_selection import cross_validate

df = pd.read_csv('gallstone.csv')

features = [
    'Gender',
    'Comorbidity',
    'Coronary Artery Disease (CAD)',
    'Hypothyroidism',
    'Hyperlipidemia',
    'Diabetes Mellitus (DM)',
    'Hepatic Fat Accumulation (HFA)'
]
X = df[features]
y = df['Gallstone Status']
```

b) Code segment:

```
dtc = tree.DecisionTreeClassifier(max_leaf_nodes=7)
dtc.fit(X, y)
print(tree.export_text(dtc, feature_names=features))
```

c) Tree:

```
|--- Hepatic Fat Accumulation (HFA) <= 1.50
|   |--- Hepatic Fat Accumulation (HFA) <= 0.50
|   |   |--- Gender <= 0.50
|   |   |   |--- class: 0
|   |   |   |--- Gender > 0.50
|   |   |       |--- Hypothyroidism <= 0.50
|   |   |       |   |--- class: 1
|   |   |       |   |--- Hypothyroidism > 0.50
|   |   |       |       |--- class: 0
|   |   |--- Hepatic Fat Accumulation (HFA) > 0.50
|   |       |--- class: 0
|   |--- Hepatic Fat Accumulation (HFA) > 1.50
|   |   |--- Gender <= 0.50
|   |   |   |--- class: 1
|   |   |   |--- Gender > 0.50
|   |       |--- Hepatic Fat Accumulation (HFA) <= 2.50
|   |       |   |--- class: 1
|   |       |   |--- Hepatic Fat Accumulation (HFA) > 2.50
|   |       |       |--- class: 0
```

d) Code segment

```
scores = cross_validate(dtc, X, y, cv=10, scoring='accuracy')

accuracies = [f"{score:.4f}" for score in scores['test_score']]
print("The accuracies:", ", ".join(accuracies))
print(f"Average: {scores['test_score'].mean():.4f}")
```

e) Accuracies:

0.7500	0.5625	0.6562	0.6562	0.5938
0.6875	0.5625	0.4688	0.4375	0.3871

Average: 0.5762